Direct and indirect proofs

1 Sari wants to prove that $(a+b)^2 \geq 2ab$ for all integers a and b.
Her proof is shown below:

$$\begin{aligned}(a+b)^2 &= a^2 + b^2 + 2ab \\ &\geq 2ab \quad \text{since } \square\end{aligned}$$

What should she write in the □ to complete her proof?

2 Prove that $(x+y)^2 - 9(x-y)^2 = 4(2x-y)(2y-x)$.

3 Use the method of completing the square to prove that $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ for all quadratics of the form $ax^2 + bx + c = 0$.

4 If N is an even integer, prove that $\dfrac{N^2}{2}$ is an even integer.

5 Prove that angle sum of a straight line is 180°.

6 Prove that the sum of two consecutive numbers is odd.

7 Prove that $(a+2)^2 - (a-2)^2$ is divisible by 8 for any positive whole number a.

8 Prove that the sum of two consecutive odd numbers is an even number.

9 Prove that the distance between two points (a, b) and (c, d) is given by $\sqrt{(c-a)^2 + (d-b)^2}$.

10 Prove that if we subtract 1 from a positive odd square number, the answer is always divisible by 8.

11 Prove that $0.\dot{1}\dot{8}$ is a rational number.